LESSON 4.5
PLOTTING, ORDERING, AND COMPARING RATIONAL NUMBERS – PART 2

LESSON INTRODUCTION

MA.6.NSO.1.1 Extend previous understanding of numbers to define rational numbers. Plot, order, and compare rational numbers.

LEARNING TARGETS

- I can plot, compare, and order rational numbers.

BENCHMARK COHERENCE

BUILDING ON

MA.5.NSO.1.4 Plot, order, and compare multidigit numbers with decimals up to the thousandths.

WORKING TOWARD

MA.8.NSO.1.1 Extend previous understanding of rational numbers to define irrational numbers within the real number system. Locate an approximate value of a numerical expression involving irrational numbers on a number line.

ESSENTIAL QUESTIONS

- How do you convert a decimal to a fraction and a fraction to a decimal?
- How can you compare rational numbers using a number line?

LESSON NARRATIVE

Students extend their learning from Lessons 3-4 to plot, compare, and order rational numbers. Students plot rational numbers on a number line. They use the numbers' placement on the number line to order and compare the numbers. Students convert fractions to decimals, and vice versa, to aid in comparing and ordering rational numbers. Students only compare negative values expressed in the same form for specified intervals along the number line. Positive values are compared in any form.

COMPONENT SUMMARY

4.5.1 WARM-UP *(~5 MIN.)*

Error analysis for plotting negative values on a number line

4.5.2 ACTIVITY *(10 MIN.)*

Place numbers on a life-sized number line

4.5.3 EXPLORATION *(10 MIN.)*

Compare unknown numbers based on their placement on a number line

4.5.4 GUIDED INSTRUCTION *(5-10 MIN.)*

Compare rational numbers and analyze different methods of comparing rational numbers

4.5.5 TRY IT *(5 MIN.)*

Practice plotting, ordering, and comparing rational numbers

4.5.6 YOUR TURN! *(5 MIN.)*

Additional practice plotting, ordering, and comparing rational numbers

4.5.7 WRAP-UP *(~5 MIN.)*

Demonstrate understanding of plotting, ordering, and comparing rational numbers

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Rational number
- Less/Greater than
- Equivalent
- Opposite

MATERIALS

- Class number line
- Sticky notes

ADDITIONAL PREPARATION

- (Optional) Print and cut out the set of rational number cards if you do not plan to use sticky notes for 4.5.2 Activity.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may interpret larger denominators as larger in value.
- Students may interpret decreasing negative values as getting larger in values as the number gets “bigger.”
- Students may believe inequality symbols can only be read left to right.

THINGS TO CONSIDER

- In 4.5.4 Guided Instruction, prompt students to reflect on the different strategies they learn for comparing and ordering rational numbers written in different forms.
- Consider using the Think Aloud strategy when demonstrating how to determine the location of a rational number on the number line.

LITERACY AND LANGUAGE ROUTINES

Collect and Display

In 4.5.3 Exploration, circulate while students are working with their partners. Collect different answers, strategies, and justifications. Debrief the activity by displaying different students’ work. Discuss the students’ responses.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

Students represent rational numbers as decimals, percentages, and fractions, (i.e.

0.4, 40%, and $\frac{2}{5}$). Students use equivalent forms to compare and order the rational numbers in 4.5.6 Your Turn!

DIFFERENTIATE INSTRUCTION

Consider utilizing On-Ramp to 6th Grade as an additional tool for scaffolding content related to conversions between fractions and decimals and percentages.

- Number Lines: Mixed Numbers
- Number Lines: Fractions
- Representing Fractions on a Number Line
- Compare and Order Decimals with Models
- Compare and Order Decimals with >, =, or <
- Comparing Fractions Using Symbols >, =, or <
- Comparing Fractions with Like Numerators and Denominators
- Compare and Order Whole Numbers with >, =, or <

4.5.1 WARM-UP: FIND THE ERROR

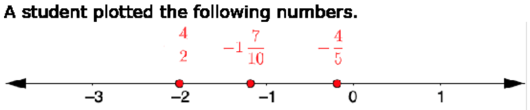
Component Overview: *Students analyze a sample of student work. They determine which values are plotted correctly and which are not. Focus on asking students to justify their answers to check for understanding.*

ENRICHMENT

- What feedback would you give to a student who made the errors you noticed?
- How can we use integers to estimate the location of fractions on a number line?
- How do you know -1 $\frac{7}{10}$ should be closer to -2 than to -1? Could we say the same thing for -$\frac{4}{5}$? Why or why not?

4.5.1 Warm-Up: Find the Error

A student plotted the following numbers.



1. Which values are plotted correctly?

   $-\frac{4}{2}$

2. Which values are NOT plotted correctly?

   $-1\frac{7}{10}$ $-\frac{4}{5}$

4.5.2 Activity: Class Number Line

Your teacher will give you a sticky note with a rational number written on the front.

Write your name on the back of the sticky note and place your number in the correct location on the class number line.

Class Number Line Values (ordered from least to greatest)

$-4\frac{3}{4}$	$-4\frac{7}{10}$	$-4\frac{1}{2}$	$-4\frac{1}{5}$	$-4\frac{1}{10}$
-3.9	-3.75	-3.7	-3.25	-3.1
$-2\frac{97}{100}$	$-2\frac{1}{5}$	$-\frac{4}{2}$	$-1\frac{7}{10}$	$-\frac{4}{5}$
$\frac{1}{4}$	36%	0.6	78%	$\frac{91}{100}$
$1\frac{1}{5}$	125%	$\frac{3}{2}$	1.9	2.01
$2\frac{1}{10}$	$2\frac{3}{5}$	2.75	3.25	$3\frac{3}{10}$

4.5.2 ACTIVITY: CLASS NUMBER LINE

Component Overview: *Construct a class number line on a wall in your classroom, the floor, or even a virtual line projected onto the board or online using a platform like Desmos. You can either write the numbers on the notes ahead of time or simply provide a list of numbers that students can then select and write on their sticky notes.*

TEACHER MOVES

The values can either be written on sticky notes, or cards could be used with tape.

DIFFERENTIATE INSTRUCTION

Consider your students when assigning the values or provide them with an opportunity to select their own value.

Consider also chunking this activity so that groups of 5 are formed and have them decide how to order within their group first, before placing their values on the class number line.

TEACHER MOVES

Probe students to share their reasoning.

- How did you determine where to place your number?
- What integer is your number closest to?
- How do you determine where to plot decimals and fractions?
- Did the placement another student made for their value help you determine where to place yours? Explain.

4.5.3 EXPLORATION: COMPARING POINTS ON A NUMBER LINE

Component Overview: *Students compare unknown numbers based on their placement on a number line. Circulate while students are working with their partners. Collect different answers, strategies, and justifications. Debrief the activity by displaying different students' work. Discuss the students' responses.*

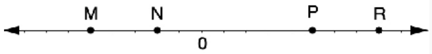
TEACHER MOVES

Think-Pair-Share

Depending on the needs of your students, consider giving students the opportunity to engage independently first and refine their responses with a partner before sharing.

4.5.3 Exploration: Comparing Points on a Number Line

A number line is shown with points M, N, P, and R. Work with your partner to answer the questions about the number line.



1. Use each of the given phrases to describe or compare the values of points M, N, P, R. Each statement will use one or two of the points.

Phrase	Description or Comparison of M, N, P or R
less than	*Answers will vary.* *M is less than N.*
opposite of	*Answers will vary.* *M is the opposite of P.*
greater than	*Answers will vary.* *R is greater than P.*
negative number	*Answers will vary.* *M is a negative number.*

2. If P is $2\frac{1}{2}$, what are the values of M, N, and R?
   M is $-2\frac{1}{2}$, *N is* -1, *R is* 4
3. If N is -0.4, what are the values of M, P, and R?
   M is -1, *P is* 1, *R is* 1.6
4. If R is 200%, what are the values of M, N, and P?
   M is -125%, *N is* -50%, *P is* 125%
5. If M is $-1\frac{2}{3}$, what are the values of N, P, and R?
   N is $-\frac{2}{3}$, *P is* $1\frac{2}{3}$, *R is* $2\frac{2}{3}$

4.5.4 GUIDED INSTRUCTION: COMPARING RATIONAL NUMBERS IN DIFFERENT FORMS

Component Overview: *Students compare fractions, decimals, and percentages using inequality symbols. Students analyze different methods of comparing rational numbers, including using a number line, finding a common denominator, and rewriting in an equivalent form.*

TEACHER MOVES

There is more practice in the next activity, so spend more time on reasoning and verbalizing strategies than finding answers (process over product).

DIFFERENTIATE INSTRUCTION

Consider having all students read all four strategies at once and set up a Four Corners named for each student. Students will read all four strategies, then walk to a corner based on the approach or strategy they align with most. At each corner, students explain why they chose that corner.

4.5.4 Guided Instruction: Comparing Rational Numbers in Different Forms

1. Use the symbols $>$, $<$, or $=$ to compare each pair of values.

   $9.02 > 9.2\%$ $\qquad -\frac{3}{5} > -\frac{13}{20}$ $\qquad 6.050 = 6.05$

   $0.4 > \frac{9}{40}$ $\qquad 2\frac{3}{4} > 2.624$ $\qquad -4\frac{1}{3} < -\frac{10}{3}$

2. Emma and Noah shared their strategy for comparing values written as fractions. Write your strategy in the box and note any **differences** from Emma's or Noah's strategy.

Emma's Strategy	Noah's Strategy
When Emma compares a negative fraction to a negative fraction, she starts by thinking about which whole numbers each fraction lies between and which each is closest to.	When Noah compares a negative fraction to a negative fraction, he uses common denominators.
Answers will vary. I converted fractions into decimals to compare the values. My strategy is different from Emma's because she estimated, and my answer is different from Noah's because he did not use decimals.	

3. Angel and Sophia share their strategies for comparing a value written as a decimal to a value written as a fraction. Write your strategy in the box and note any **similarities** from Angel's or Sophia's strategy.

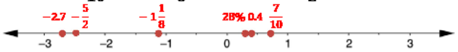
Angel's Strategy	Sophia's Strategy
When Angel compares a positive decimal to a positive fraction, he rewrites the fraction as an equivalent decimal.	When Sophia compares a positive decimal to a positive fraction, she plots them on a number line.
Answers will vary. I converted fractions into decimals to compare the values, so my strategy was the same as Angel's.	

4.5.5 Try It: Plotting, Ordering, and Comparing Rational numbers in Different Forms

1.

a. Plot each value on the number line. Label each point with its numeric value.

-2.7 $\quad \frac{7}{10}$ $\quad -1\frac{1}{8}$ $\quad 0.4$ $\quad -\frac{5}{2}$ $\quad 28\%$



b. What strategies did you use that were helpful when plotting the rational numbers written in different forms on the number line?

Answers will vary. I converted all the numbers to decimals to help me plot them on the number line.

c. Order the values from greatest to least.

$\frac{7}{10}$ $\quad 0.4$ $\quad 28\%$ $\quad -1\frac{1}{8}$ $\quad -\frac{5}{2}$ $\quad -2.7$

d. Complete the inequality statement to show the relationship between the values plotted on the number line.

$-2.7 < -\frac{5}{2} < -1\frac{1}{8} < 28\% < 0.4 < \frac{7}{10}$

4.5.5 TRY IT: PLOTTING, ORDERING, AND COMPARING RATIONAL NUMBERS IN DIFFERENT FORMS

Component Overview: *Students plot, order, and compare rational numbers in a variety of forms. The questions include fractions, decimals, and percentages. Encourage students to choose one or more strategies based on the previous component.*

ENRICHMENT

How does the order of values, from greatest to least, relate to the location on a number line? Is this always true? Why or why not?

4.5.6 YOUR TURN!

Component Overview: *Students continue practicing plotting, ordering, and comparing rational numbers.*

ENRICHMENT

Students might still struggle interpreting negative values or fractions in order to compare. Consider what question stems you can provide for students to use when they get stuck.

- What whole number is this value closest to?
- Is it greater than or less than that whole number?
- How do you know? Give an example.
- Should that value be plotted on the left or the right of that whole number?
- Is that value greater or less than…?

4.5.6 Your Turn!

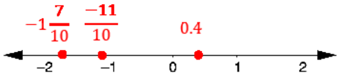
1. Order the values from least to greatest.

$\frac{1}{2}$ $\quad 0 \quad 1 \quad -1\frac{1}{2} \quad -\frac{1}{2} \quad -1$

$-1\frac{1}{2} \quad -1 \quad -\frac{1}{2} \quad 0 \quad \frac{1}{2} \quad 1$

2. Plot the values on the number line. Label each point with its numeric value.

$0.4 \quad -1\frac{7}{10} \quad -\frac{11}{10}$



3. Order the values from greatest to least.

$-2 \quad 0 \quad -\frac{1}{6} \quad 1\frac{3}{8} \quad -3\frac{1}{3} \quad 1.4 \quad \frac{5}{4} \quad -\frac{13}{10} \quad \frac{21}{5}$

$\frac{21}{5} \quad 1.4 \quad 1\frac{3}{8} \quad \frac{5}{4} \quad 0 \quad -\frac{1}{6} \quad -\frac{13}{10} \quad -2 \quad -3\frac{1}{3}$

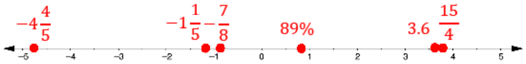
4. Use the symbols $>$, $<$, or $=$ to compare each pair of values.

$8.7 > 8.689$ $\qquad -\frac{5}{3} < -\frac{7}{10}$ $\qquad 7.2 < 7\frac{1}{3}$

$0.38 > 5\%$ $\qquad -4\frac{5}{8} < -4.5$ $\qquad -3\frac{2}{5} = -3.4$

4.5.7 Wrap-Up: Ticket Out the Door

1. Use the values 3.6, $-4\frac{4}{5}$, 0.7, $-1\frac{1}{5}$, $\frac{15}{4}$, 89%, and $-\frac{7}{8}$. Plot the values on the number line. Label each point with its numeric value.



2. List the values 3.6, $-4\frac{4}{5}$, 0.7, $-1\frac{1}{5}$, $\frac{15}{4}$, 89%, and $-\frac{7}{8}$ from least to greatest.

   $-4\frac{4}{5}$ $-1\frac{1}{5}$ $-\frac{7}{8}$ 0.7 89% 3.6 $\frac{15}{4}$

3. Use the symbols <, >, or = to compare each pair of values.

   $-12.742 > -12.9$ $1\frac{2}{9} < 150\%$ $3\frac{7}{12} < 3.7$

4.5.7 WRAP-UP: TICKET OUT THE DOOR

Component Overview: *In this formative assessment, students demonstrate their ability to plot, order, and compare rational numbers. Use the data gathered to make instructional decisions regarding remediation prior to the next lesson.*

DIFFERENTIATE INSTRUCTION

By plotting values in question 1, students must use skills of comparing and ordering they would need in question 2. Question 3 will be important in gathering data for students' understanding of inequality symbols.